

Homework 6 - Scrum Basics

Sprint 0: Project Setup & Planning

Sprint Goal:

Define project scope and structure by planning and organizing upcoming sprints.

Sprint Scope:

- Define overall project vision and objectives
- Identify target users (Doctor, Pharmacist, Patient)
- Plan and structure Sprint 1, Sprint 2, and Sprint 3
- Define key features (patient management, prescriptions, stock, alerts)
- Select tools (Figma for prototype, React & NextJS for development)
- Define workflow and responsibilities

Planned Sprints:

- **Sprint 1:** Personas, Scenarios and User Stories Definition
- **Sprint 2:** Validation and Usability Testing
- **Sprint 3:** Prototype Improvement and System Implementation

Key Deliverables:

- Defined project scope and goals
- Sprint roadmap (3 sprints planned)
- Initial feature list
- Selected tools and technologies
- Team roles and workflow defined

Acceptance Criteria:

- Project goals clearly defined
- All future sprints planned and structured
- Core features identified
- Tools and technologies selected
- Team understands workflow and next steps

Sprint 1: Personas, Scenarios & User Stories Definition

Sprint Goal:

Define clear personas, identify key problem scenarios and translate them into structured user stories with acceptance criteria.

Sprint Scope:

Personas:

- Doctor (Dr. Vangel)
- Pharmacist (Lea)
- International Student (Elena)

Problem Scenarios:

- Adding a new patient
- Patient search and record access
- Prescription updates and synchronization
- Searching for a patient to dispense medication
- Handling a prescription update from a doctor
- Seeking Medical Attention Abroad
- Emergency Care While Traveling

User Stories:

- Patient search by unique ID
- Digital prescription creation
- Update diagnosis and treatment
- Prescription dispensing and stock update
- Alerts for low stock and expiry
- View patient record

Key Deliverables:

- 3 fully defined personas
- Documented problem scenarios for each persona
- Structured user stories for all scenarios
- Acceptance criteria for each user story

Acceptance Criteria:

- Each persona includes goals, frustrations and behaviors
- Each scenario clearly describes the problem and current limitations
- Each user story follows format: *As a [role], I want [goal], so that [benefit]*
- Each user story includes clear, testable acceptance criteria

Sprint 2: Validation & Usability Testing

Sprint Goal:

Validate system assumptions through user feedback and improve usability of core features.

Sprint Scope:

Hypothesis Testing:

- Validate need for centralized patient records (doctor)
- Validate real-time prescriptions and stock updates (pharmacist)
- Validate cross-border access to medical data (patient)
- Identify improvements (timestamps, sync status, highlighted changes, allergy visibility)

Usability Testing:

- Test core flows: Add Patient, Search Patient and View Medical Record, Check the prescription and mark it as fulfilled, Find Low Stock Alerts and find one low-stock and one near-expiry medicine, Find which prescriptions are In Progress.
- Measure task completion time and success
- Identify usability issues and confusing steps

Key Deliverables:

- Validated hypotheses with interview insights
- Usability test results and performance metrics
- Identified usability issues (navigation, feedback, clarity)
- List of improvements for next iteration

Key Findings:

- Problem is confirmed for all personas
- System is generally easy to use
- Main issues:
 - Add Patient flow clarity
 - Prescription completion flow complexity
 - Need for better feedback and visibility

Acceptance Criteria:

- All hypotheses tested with real users expect the one private pharmacist
- All core tasks completed successfully by participants
- Usability issues clearly documented
- Actionable improvements defined for next sprint

Sprint 3: Prototype Improvement & System Implementation

Sprint Goal:

Improve the prototype based on usability feedback and start implementing core functionalities with a real database.

Sprint Scope:

UI/UX Improvements (from Sprint 2 feedback):

- Improve *Add Patient* flow clarity
- Simplify prescription completion flow
- Add clear confirmation messages (success/error)
- Highlight allergies and critical patient data
- Show last update timestamp and author
- Indicate sync status (real-time / pending)
- Improve navigation and feature visibility

Functional Development:

- Implement patient registration with validation (no duplicates)
- Implement patient search by unique ID

- Implement digital prescription creation and updates
- Implement prescription status (In Progress / Completed)
- Implement stock updates after dispensing
- Implement alerts (low stock and near expiry)

Database Integration:

- Design and connect centralized database
- Store patient records, prescriptions and stock data
- Enable real-time data synchronization (basic level)

Key Deliverables:

- Updated interactive prototype (improved UX)
- Working backend with database connection
- Core features functional (patient, prescription, stock)
- Basic real-time data handling

Acceptance Criteria:

- Users can add and find patients without confusion
- No duplicate patient entries allowed
- Prescriptions update and sync correctly
- Stock updates automatically after dispensing
- Alerts display accurate data
- UI provides clear feedback and critical info visibility